Milestone 1 Report

CSCE3701 - Software Engineering (Fall 2025)

Group Number: 1

Sections:-

CSCE3701 - Software Engineering (Fall 2025)

Dr. Yasmine Afify

Group Number: 1

Milestone 1 Report

Overview of the SW

The Hospital Management System (HMS) is a software solution developed to enhance the efficiency and accuracy of hospital operations. Instead of relying on traditional, paper-based systems, the HMS integrates various hospital functions such as patient registration, appointment scheduling, staff management, and billing into a single, secure application.

The Hospital Management System supports several types of users, including patients, doctors, nurses, administrative staff, and system administrators. Patients can create and manage their own accounts, book and cancel appointments, view their medical records, and track billing information. Doctors and nurses can access patient medical histories, update treatment details, and manage their daily schedules. Administrative staff members can handle appointment coordination, shifting schedule, patient admissions, billing, and financial records. Meanwhile, system administrators could maintain data security, control user access, and oversee the overall performance of the system.

The motivation behind developing the Hospital Management System is to overcome the inefficiencies and challenges that hospitals face when using manual methods. Many healthcare facilities in Egypt struggle with maintaining accurate records, long patient wait times, and difficulties in communication between departments. By automating these tasks, the HMS minimizes human error, reduces administrative workload, and provides real-time data access for making better decisions. In addition, it improves patient experience by allowing them to interact with hospital services online without the need for physical visits.

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The product backlog

User Story #1

As a patient, I want to book appointments so that I can visit my doctor at a convenient time.

Acceptance Tests:-

- Given that the patient is logged into the patient portal, when they navigate to the appointment booking page, then they can view the list of hospital clinics to choose from.
- Given that the patient is in the booking appointment page, when they select a clinic, then the system displays all doctors and their specialties in that clinic.
- Given that the patient has selected a doctor, when they open the doctor's schedule, then all available appointment time slots are displayed.
- Given the patient selects a time slot, when they confirm the booking and the booking is successfully processed, then the appointment is added to the patient's upcoming appointments, and a confirmation email or notification is sent to the patient.
- Given that the doctor's schedule is linked to the booking system, when a patient books an appointment, then that time slot becomes unavailable for other patients.

User Story #2

As a patient, I want to cancel my booked appointments so that I can free up the slot for others.

Acceptance Tests:-

- Given that the patient views their appointment list, when they click the "Cancel" button for a specific appointment, then a confirmation prompt should appear asking them to confirm the cancellation.
- Given that the patient confirms the cancellation, when the system processes the request, then the appointment should be removed from the patient's appointment list, and the doctor's schedule should be updated to free the time slot.
- Given that the appointment has been successfully canceled, when the cancellation is completed, then the system should display or send a confirmation message (on-screen and/or via email).

Functional Requirements

- The system shall allow a logged-in patient to view a list of all their booked appointments.
- The system shall provide a “Cancel” button next to each appointment.
- The system shall remove the canceled appointment from the patient’s appointment list.
- The system shall update the doctor’s schedule to free the corresponding time slot.
- The system shall display an on-screen confirmation message and/or send an email notification confirming the successful cancellation.
- The system shall log the cancellation event for tracking purposes.

Non-Functional Requirements

- The cancellation should be processed within 3 seconds after clicking cancel.
- The system should ensure that both the patient’s and the doctor’s records remain consistent after cancellation.
- Only authenticated and authorized patients can cancel their own appointments.
- The cancellation feature should be available 24/7.

User Story #3

As a patient, I want to reschedule my appointment so that I can change the time without losing my booking.

Acceptance Tests:-

- Given that a patient is logged into the system, when they access the “My Appointments” page, then the system should display an option to reschedule an appointment.
- Given that the patient chooses to reschedule an appointment, when they select a new available date and time slot, then the system should remove the old appointment and create a new one for the selected slot in the appointment page, and send a confirmation email/message.
- Given that the appointment time has changed, when the reschedule is completed, then the doctor’s schedule should also be updated to reflect the new and freed time slots.

Functional Requirements

- The system shall allow a logged-in patient to view all current appointments.
- The system shall display a “Reschedule” option next to each upcoming appointment.
- The system shall show the available time slots for booking a new appointment.
- The system shall allow the patient to select a new date and time slot.
- The system shall cancel the old appointment and create a new one for the chosen slot.
- The system shall update the doctor’s schedule accordingly.
- The system shall send a confirmation message/email after successful rescheduling.
- The system shall keep a record of reschedule history for tracking.

Non-Functional Requirements

- The system should complete the rescheduling process within 5 seconds.
- The date/time picker should be intuitive and prevent the selection of unavailable slots.
- The system must prevent overlapping appointments for both patient and doctor.
- Only the patient who booked the appointment can reschedule it.
- The system should handle multiple concurrent rescheduling requests without errors.
- The patient and doctor must both receive instant notifications after the change.

User Story #4

As a patient, I want to view my upcoming and past appointments so that I can track my visit history.

Acceptance Tests:-

- Given that a patient is logged into the system, when they access the “My Appointments” page, then the system should display upcoming and past appointments in separate sections.
- Given that the appointment list is displayed, when the patient selects a certain appointment, then it should show the appointment date, time, and doctor’s name.
- Given that a patient has canceled an appointment, when the appointment list loads, then the canceled appointment should appear with a “Canceled” status label.

- Given that appointment data changes (e.g., new booking, cancellation, or reschedule), when the patient refreshes or reopens the page, then the appointment list should update automatically to reflect the latest information.

Functional Requirements

- The system shall allow a logged-in patient to view all appointments (past, upcoming, and canceled).
- The system shall display upcoming and past appointments in separate sections.
- The system shall display details such as appointment date, time, and doctor's name.
- The system shall show a "Canceled" status label for any canceled appointment.
- The system shall refresh or reload automatically to reflect any new, canceled, or rescheduled appointments.
- The system shall allow sorting and filtering appointments by date or doctor's name.

Non-Functional Requirements

- The appointment list should load within 2 seconds of accessing the page.
- The interface should clearly separate upcoming, past, and canceled appointments.
- The displayed data must always reflect the latest state of the database.
- The appointment list should be viewable on both desktop and mobile screens.
- Only the authenticated patient can view their appointment history.

User Story #6

As a doctor, I want to view all appointments scheduled with me so that I can prepare accordingly.

Acceptance Tests:-

- Given that one or more patients have booked appointments with me, when I log in, then the appointments appear in my schedule.
- Given that no patients have booked appointments with me, when I log in, then no appointments appear in my schedule and a message appears saying "You have no appointments".

- Given that I have a scheduled appointment, when I press on an appointment, then the patient information appears to me.

User Story #7

As a receptionist, I want to view all appointments across doctors so that I can manage the clinic schedule.

Acceptance Tests:-

- Given that there are appointments made for today, when I log in, then I see the appointment made in chronological order.
- Given that there is an appointment made for today, when I open the dashboard, then I can see the clinic number, doctor and patient's names.
- Given that there are no appointments made for today, when I log in, then an appropriate message appears.

User Story #8

As a patient, I want to create a new account by providing my personal details and contact information so that I can access the hospital's online services securely.

Acceptance Tests:-

- Given that the patient is on the signup page, when they provide all required fields (name, email, phone, password), then the system should create a new account.
- Given that any required field is missing, when the patient clicks "Submit," then the system should display an error message
- Given that the account is created, when the signup is successful, then the system should send a verification email.

Functional Requirements (FRs):

1. The system shall provide a registration form for patients to create a new account.
2. The system shall allow patients to enter personal details (e.g., full name, date of birth, gender) and contact information (e.g., email, phone number).

3. The system shall validate all required fields for correctness and completeness like valid email format, mandatory fields not empty.
4. The system shall ensure that the email and/or phone number is unique and not already registered.
5. The system shall allow patients to create a secure password and enforce password strength rules (e.g., minimum length, special characters).
6. The system shall store patient information securely in the database.

Non-Functional Requirements (NFRs):

1. All personal and contact information must be transmitted and stored securely (e.g., HTTPS, encrypted database storage).
2. The registration form must be user-friendly, clear, and easy to complete.
3. The system should process the registration request within 6 seconds under normal load.
4. The system must reliably store patient information and prevent data loss.

User Story #9

As a system user, I want to access a login page where I can enter my email and password so that I can securely sign in to my account.

Acceptance Tests:-

- Given that the patient has a verified account, when valid credentials are entered, then the system should log them in successfully.
- Given that having invalid credentials, when the login button is pressed, then an "invalid email or password" message should appear.
- Given that the patient is logged in, when they close and reopen the browser, then they should stay logged in if the "Remember me" option was selected.

Functional Requirements (FRs):

1- The system shall authenticate the users by checking their email and password in the stored records in the database.

2- The system shall display an error message "Invalid email or password" if the email and the password do not match any registered account.
3- After verifying the email and password successfully, the system shall redirect the patient to the application dashboard displaying their personalized information and available hospital services.

Non-Functional Requirements (NFRs):

1- The system shall complete the login process and load the dashboard within 3 seconds under normal network conditions.

2- The system shall store passwords using encryption and hashing algorithms.

User Story #10

As a system user, I want to use a "Forgot Password" feature so that I can reset my password if I forget it and regain access to my account.

Acceptance Tests:-

- Given the patient forgets their password, when they click "Forgot Password," then the system should ask for their registered email.
- Given that the email is valid, when they submit it, then the system should send a password reset link.
- Given that the link is clicked, when the new password is entered, then the system should update it successfully.

Functional Requirements (FRs):

1- The system shall provide a "Forgot Password" option on the login page.

2- The system shall allow the patient to enter their registered email or username to request a password reset.
3- The system shall verify that the entered email or username exists in the patient database.
4- The system shall send a password reset link or code to the patient's registered email.
5- The system shall allow the patient to enter a new password via the reset link or code.

6- The system shall validate that the new password meets the security requirements (e.g., minimum length, special characters.

7- The system shall update the patient's account with the new password after successful reset.
8- The system shall notify the patient if the password reset was successful.

Non-Functional Requirements (NFRs):

1- The password reset process must use secure protocols (e.g., HTTPS) to protect sensitive data.

2- The password reset email should be sent within 1 minute of the request.

3- The reset process should be simple and clear for patients to follow.

4- The system must ensure the reset link or code is valid for a limited time (e.g., 15–30 minutes).

User Story #11

As a patient, I want to receive an authentication email when I complete the signup process so that my account can be verified and activated securely.

Acceptance Tests:-

- Given that the patient successfully signs up, when the registration completes, then the system should automatically send a verification email.
- Given that the email is received, when the patient clicks the link, then the system should mark their account as verified.
- Given that the account is not verified, when they attempt to login, then an error message will appear stating that they should verify first.

Functional Requirements (FRs):

1. The system shall provide a signup form for patients to create an account.
2. The system shall validate that all required signup fields (e.g., email, password) are correctly filled.
3. Upon successful signup, the system shall generate a unique authentication token or code for the patient.
4. The system shall send an authentication email containing the verification link or code to the patient's registered email address.
5. The system shall allow the patient to click the verification link or enter the code to activate their account.

6. The system shall mark the patient's account as verified and active after successful authentication.
7. The system shall notify the patient if the account verification is successful.

Non-Functional Requirements (NFRs):

1. The authentication email and activation link must be transmitted using secure protocols (e.g., HTTPS).
2. The verification email should be sent within 1 minute of completing the signup process.
3. The email content and verification process should be simple and easy for patients to follow.
4. The verification token or link should expire after a set period of 24 hours for security purposes.

User Story #12

As a nurse, I want to view my assigned shifts on a schedule page so that I can easily keep track of my work hours.

Acceptance Tests:-

- Given the nurse is logged in, when they navigate to the "My Schedule" page, then their assigned shifts (date, time, department) are displayed.
- Given the nurse has no assigned shifts, when they open the "My Schedule" page, then a message appears stating "No shifts assigned yet."
- Given the nurse is viewing their schedule, when they scroll or navigate through dates, then shifts are displayed in chronological or calendar order.

User Story #13

As an admin, I want to create and manage available shifts so that I can assign them to nurses later.

Acceptance Tests:-

- Given the admin is logged in, when they open the "Manage Shifts" page and create a new shift (with date, time, department), then the new shift is saved and appears in the shift list.

- Given the admin views an existing shift, when they edit or delete it, then the system updates or removes the shift accordingly.
- Given the admin is viewing the list of shifts, when they look at each entry, then each shift's status (Unassigned or Assigned) is clearly shown.

User Story #14

As a nurse, I want to mark my unavailable days so that I'm not assigned shifts then.

Acceptance Tests:-

- Given the nurse is logged in, when they mark specific days as unavailable, then those days appear visually distinct on their calendar.
- Given a nurse has marked a date as unavailable, when the admin tries to assign them a shift on that date, then the system prevents the assignment and displays an error message.

User Story #15

As an admin, I want to assign nurses to available shifts so that each nurse can view their work schedule.

Acceptance Tests:-

- Given there are unassigned shifts and available nurses, when the admin selects a shift and assigns a nurse, then the shift's status changes to Assigned and appears in the nurse's schedule.
- Given the nurse is already assigned to another shift at the same time, when the admin tries to assign them to a new one, then the system prevents the conflict and displays an alert.
- Given a nurse is assigned a new shift, when the assignment is saved, then the nurse is notified or sees the update immediately on their schedule.

User Story #16

As a patient, I want to send messages or inquiries to hospital support so that I can get answers without visiting in person.

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Acceptance Tests:-

- Given the patient is logged into the portal, when they navigate to the “Contact Hospital” or “Messages” section, then they should see an option to compose and send a new message.
- Given that the patient sends a message through the portal, when the message is submitted, then the system should securely store it with a timestamp.
- Given that the patient has successfully sent a message, when the system confirms submission, then the patient should receive an on-screen confirmation that the message was sent.

Functional Requirements

- The system shall allow only logged-in patients to access the “Contact Hospital” or “Messages” section.
- The system shall provide a form interface where patients can compose a new message or inquiry.
- The system shall allow patients to include a subject, message body, and optional attachment.
- The system shall store each message securely in the database with a timestamp and sender ID.
- The system shall display an on-screen confirmation message after the message is successfully sent.
- The system shall allow patients to view previously sent messages and their status.
- The system shall notify support staff when a new message is received.

Non-Functional Requirements

- Messages and patient data must be encrypted.
- Sending or viewing messages should complete within 3 seconds.
- The message form should have a simple, user-friendly design with validation for required fields.
- The messaging feature should be accessible 24/7.

- Only authorized hospital staff can view patient messages.
- Messages must not be lost even if the system experiences downtime.

User Story #17

As a support group member, I want to view and reply to patient messages so that I can provide timely assistance and address patient concerns.

Acceptance Tests:-

- Given that a patient has sent a message through the portal, when the support group logs into their staff portal, then they should be able to see the new message in their inbox.
- Given that the support group opens a patient's message, when they type a reply and click "Send," then the system should securely deliver the response to the patient's portal.
- Given that a reply is sent, when the message is successfully delivered, then the patient should receive a notification confirming that a response is available.

Functional Requirements

- The system shall allow only authenticated support group members to access the support portal inbox.
- The system shall display all incoming patient messages in a structured inbox with message ID, sender name, subject, and timestamp.
- The system shall allow support members to open, read, and reply to each message.
- The system shall allow support group members to send replies securely to the patient's portal and store them in the database.
- The system shall send a notification to the patient when a reply is received.
- The system shall track the status of messages (e.g., "Unanswered," "Replied").

Non-Functional Requirements

- Only authorized support staff should have access to patient communications.
- Replies must not contain sensitive patient data unless necessary and protected by encryption.
- Messages and replies should load within 2 seconds of request.

- Replies must always securely reach the patient portal.
- The system should handle simultaneous replies from multiple support staff.
- The interface should clearly mark unread messages and provide a rich text editor for responses.

User Story #18

As a patient, I want to provide feedback or rate my experience after an appointment so that the hospital can improve its services.

Acceptance Tests:-

- Given that the patient has completed an appointment, when they log into the patient portal, then they should be able to see an option to provide feedback or rate their experience.
- Given that the patient submits feedback or a rating, when the feedback is sent, then the system should save it securely and show a confirmation message.
- Given that the feedback is stored, when hospital support staff access the feedback system, then they should be able to view the patient's comments and rating.

Functional Requirements

- The system shall allow patients who have completed an appointment to access a feedback or rating option.
- The system shall display a feedback form with a rating and an optional comment box.
- The system shall store feedback in the database with appointment and patient details.
- The system shall show an on-screen confirmation message once feedback is successfully submitted.
- The system shall allow hospital support or admin staff to view patient feedback and ratings through their portal.
- The system shall aggregate feedback results for reporting and quality improvement.
- The system shall prevent a patient from submitting duplicate feedback for the same appointment.

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Non-Functional Requirements

- The feedback form must be simple and accessible on both desktop and mobile.
- Submitting feedback should take no more than 2 seconds.
- Feedback data must be protected from unauthorized access and linked to the correct patient record.
- The feedback system must remain available at least 99% of the time.
- Feedback should be stored securely in the database.
- Feedback must not display patient identity to unauthorized viewers.
- The system must support a large number of feedback submissions after busy hospital days.

User Story #19

As a patient, I want to pay my hospital fees through the app so that the process is convenient.

Acceptance Tests:-

- Given that the patient has an outstanding hospital bill, when they proceed to the payment section in the patient portal, then they should be able to see multiple payment methods (using credit card, mobile wallet, or cash).
- Given that the patient selects a payment method, when they confirm their choice, then the system should record the selected payment method for that transaction.
- Given that the payment method is saved, when the patient reviews their payment summary, then they should see the chosen payment method displayed clearly.

Functional Rquirments:

- The system shall allow patients with outstanding bills to view their total due amount.
- The system shall display multiple payment methods (credit card, mobile wallet, cash).
- The system shall allow patients to select a preferred payment method.
- The system shall record and store the chosen payment method for each transaction.
- The system shall display a payment summary including the selected payment method.
- The system shall validate the payment method before confirming the transaction.

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Non-Functional Requirments:

- The payment summary should load within 3 seconds.
- Payment steps should be intuitive, with clear navigation and feedback messages.
- All payment data must be transmitted using secure (HTTPS) protocols
- The payment process should have transaction integrity — no duplicate or partial payments.
- Each payment transaction must be logged with timestamp and patient ID.

User Story #20

As a doctor, I want to search for a patient's medical record by ID, name, or national number so that I can quickly access the correct file.

Acceptance Tests:-

- Given that the user enters a patient ID, name, or national number, when they perform the search, then the system should display all matching patients with basic details (name, age, gender, ID).
- Given that multiple patients match the search input, when results are displayed, then all matching patients should appear for selection.
- Given that the user enters an invalid or empty search query, when the search is executed, then the system should display a "No results found" message.
- Given that the user performs a valid search, when the query is processed, then results should load within 3 seconds.
- Given that an unauthorized user attempts to perform a search, when they try to submit the query, then the system should display an "Access Denied" error.
- Given that a search query is executed, when results are retrieved, then the system should log the user ID and timestamp of the search.

Functional Rquirments:

- The system shall allow searches by patient ID, name, or national number.
- The system shall display all matching patients with basic details (name, age, gender, ID).

- The system shall show all matching results when multiple patients fit the search criteria.
- The system shall display a “No results found” message for invalid or empty queries.
- The system shall restrict access to authorized users (doctor role).
- The system shall log search activity (user ID, timestamp, search query).

Non-Functional Requirments:

- Search results must appear within 3 seconds for valid queries.
- Access control must ensure only authorized personnel can perform patient searches.
- The search feature must be accessible 24/7 with ≥99% uptime.
- The search function should handle concurrent requests from multiple users efficiently.
- All search actions must be recorded for traceability.
- Search interface must support partial and case-insensitive matches for convenience.

User Story #21

As a nurse, I want to view the patient’s basic information (age, gender, allergies, blood type) so I can prepare appropriate care.

Acceptance Test:-

- Given that the nurse has the required permissions, when they open a patient’s record, then the system should display accurate demographic details (age, gender, allergies, blood type).
- Given that the nurse does not have sufficient authorization, when they attempt to view sensitive data, then the system should mask or hide restricted fields.

Functional Rquirments:

- he system shall allow authorized nurses to view patient demographic information (age, gender, allergies, blood type).
- The system shall restrict access and mask or hide sensitive data when the nurse lacks sufficient authorization.
- The system shall validate user permissions before displaying patient details.

Non-Functional Requirments:

- Role-based access control (RBAC) must protect patient information.

- Patient details should load within 2 seconds of request.
- The interface must present information in a clear, readable format suitable for quick reference.
- The displayed data must always be current and consistent with the master record.
- The system should handle simultaneous access without data corruption.

User Story #22

As a doctor, I want to view the patient's medical history, including diagnoses, past visits, and treatments, to make informed medical decisions.

Acceptance Criteria:-

- Given that the doctor opens a patient's record, when they view medical history, then all past visits should appear in reverse chronological order.
- Given that a visit record is displayed, when viewed, then it should include the date, department, doctor's name, diagnosis summary, and treatment notes.
- Given that the doctor is reviewing medical history, when interacting with the interface, then they should not be able to edit or modify any data.
- Given that the patient has no recorded history, when the doctor opens the history tab, then the system should display "No medical history found."
- Given that the doctor retrieves a patient's history, when data is loaded, then it should appear within 3 seconds for standard patient profiles.
- Given that the doctor uses filters, when they select a date range or department, then the system should display only matching records.
- Given that a user without "Doctor" or "Specialist" role attempts to access history, when they try to open it, then the system should display an "Access Denied" message.

Functional Requirements

- The system shall allow doctors to view patients' medical history, including diagnoses, visits, and treatments.

- The system shall display visit records in reverse chronological order.
- Each visit record shall include date, department, doctor’s name, diagnosis summary, and treatment notes.
- The system shall restrict editing — medical history should be view-only.
- The system shall display “No medical history found” if none exists.
- The system shall allow filtering of records by date range or department.
- The system shall restrict access to authorized roles (“Doctor” or “Specialist”).

Non-Functional Requirements

- History data should load within 3 seconds for standard patient profiles.
- Only authorized users can access medical history (RBAC enforcement).
- Information should be presented in a structured, readable format (tables or collapsible sections).
- The system should ensure historical data integrity (no loss or alteration).
- Each history access should be logged with doctor ID and timestamp.
- Must support large datasets of historical records efficiently.

User Story #23

As a nurse, I want to view current medications and prescriptions so I can administer the correct doses.

Acceptance Criteria:-

- Given that the nurse opens the medication list, when the system retrieves prescriptions, then only active (status = “Ongoing”) ones should be displayed.
- Given that a prescription entry is shown, when viewed, then it should include the drug name, dosage, route, and frequency.
- Given that discontinued or expired medications exist, when the nurse opens the list, then they should be hidden by default but viewable through a filter.

- Given that a doctor updates a prescription, when the nurse refreshes the view, then the list should automatically reflect the latest data.
- Given that the nurse lacks permission, when attempting to view prescriptions, then the system should display an “Insufficient permissions” message.
- Given that the nurse views medication details, when displayed, then data should appear clearly formatted for bedside reference.

Functional Requirements

- The system shall display only active prescriptions (status = “Ongoing”) by default.
- Each prescription entry shall include drug name, dosage, route, and frequency.
- The system shall allow viewing of discontinued or expired medications via filter options.
- The system shall automatically update the displayed list when a doctor modifies a prescription.
- The system shall restrict access to authorized nurses.
- The system shall display “Insufficient permissions” when unauthorized access is attempted.

Non-Functional Requirements

- Performance: Medication lists should refresh within 2 seconds of updates.
- Usability: Medication details must be clearly formatted for bedside reference (large font, clean layout).
- Security: Prescription data should be visible only to authenticated users with proper roles.
- Reliability: Data synchronization between doctor updates and nurse view must be consistent.
- Maintainability: The system should support easy updates to medication data structures.
- Availability: The medication list feature should be available 24/7 to nursing staff.

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User Story #24

As a healthcare organization or system administrator, I want to ensure that patient medical records are stored and accessed securely, so that patient privacy is protected and sensitive information is not disclosed to unauthorized users.

Acceptance Criteria:-

- Given that a user attempts to access patient records, when they log in with valid credentials, then the system should grant access only if their role is authorized.
- Given that users have different roles, when accessing records, then their privileges should match their role type (e.g., doctor, nurse, admin).
- Given that an unauthorized user tries to open a patient record, when they make the request, then the system should deny access and display "Access Denied."
- Given that a user views or modifies patient data, when the action occurs, then it should be logged with user ID, timestamp, and action type.
- Given that data is transmitted or stored, when the process occurs, then it must be encrypted both in transit and at rest.

User Story #25

As a doctor, I want to set my weekly working hours and availability so patients can book appointments during those times.

Acceptance Criteria:-

- Given that the user has the "Doctor" role, when they open the schedule page, then they should be able to create or modify their own availability.
- Given that an unauthorized role accesses the scheduling page, when they attempt to open it, then the system should display "Access Denied."
- Given that the doctor selects available days, when they define time slots, then the system should validate that start time is earlier than end time.
- Given that the doctor saves their schedule, when validation passes, then the data should be stored and a confirmation message displayed.

- Given that a doctor has saved their schedule, when patients book appointments, then the booking system should only allow reservations within those available slots.

User Story #26

As a doctor, I want to update or cancel my schedule if my availability changes, so the hospital can adjust appointments accordingly.

Acceptance Criteria:-

- Given that the doctor is logged in, when they access their own schedule, then they should be able to edit or cancel it.
- Given that the doctor modifies a slot, when changes are saved, then the system should validate that no overlaps or invalid ranges exist.
- Given that the doctor cancels a time slot or day, when confirmed, then linked patient appointments should be marked “Cancelled by doctor” and notify affected patients.
- Given that the doctor confirms an update or cancellation, when the action is completed, then a “Schedule updated successfully” message should appear.
- Given that the doctor attempts to modify a past date, when they save, then the system should block the update and display an error.

User Story #27

As an admin/scheduler, I want to view all doctors’ schedules to manage department coverage and assign shifts efficiently.

Acceptance Criteria:-

- Given that the user has the “Admin” or “Scheduler” role, when they open the schedule view, then they should see all doctors’ schedules.
- Given that a doctor logs in, when viewing schedules, then they should only see their own.
- Given that the admin/scheduler opens the calendar view, when data is displayed, then schedules should be viewable by department, day, week, or month.

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- **Functional Requirements (FRs) and Non-Functional Requirements (NFRs):**

User Story #1

As a patient, I want to book appointments so that I can visit my doctor at a convenient time.

Functional Requirements (FRs):

1. The system shall allow patients to open the appointment booking page to view all available hospital clinics and select the desired clinic for booking.
2. The system shall enable patients to select a doctor within the chosen clinic and choose an available date and time slot for the appointment.
3. The system shall save the confirmed appointment details in the patient's and doctor's schedules in the database.
4. The system shall send a confirmation message or email to the patient after the successful booking.

Non-Functional Requirements (NFRs):

1. The system shall process and confirm an appointment booking within 3 seconds after the patient submits the booking request.
2. The system shall handle multiple concurrent appointment booking requests without performance degradation or system failure.

User Story #9

As a patient, I want to access a login page where I can enter my email and password so that I can securely sign in to my account.

Functional Requirements (FRs):

1- The system shall authenticate the users by checking their email and password in the stored records in the database.
2- The system shall display an error message "Invalid email or password" if the email and the password do not match any registered account.
3- After verifying the email and password successfully, the system shall redirect the patient to the application dashboard displaying their personalized information and available hospital services.

Non-Functional Requirements (NFRs):

1- The system shall complete the login process and load the dashboard within 3 seconds under normal network conditions.

2- The system shall store passwords using encryption and hashing algorithms.

User Story #15

As an admin, I want to assign nurses to available shifts so that each nurse can view their work schedule.

Functional Requirements (FRs):

1- The system shall allow the admin to assign one or more nurses to an existing, unassigned shift.

2- The system shall prevent the assignment of a nurse to overlapping or unavailable shifts.

3- The system shall update the nurse's schedule immediately after a successful assignment or reassignment.

Non-Functional Requirements (NFRs):

1- The system shall update and display the nurse's new shift assignment within **2 seconds** of the admin's confirmation.

2- The assignment shall be saved even if the network connection is briefly lost.

User Story #19

As a patient, I want to pay my hospital fees through the app so that the process is convenient.

Functional Requirements (FRs):

1- The system shall allow the patient to view the bills' details before payment method.

2- The system shall allow the patient to choose the payment method according to their preferences.

3- The system shall display a successful payment confirmation receipt upon successful payment.

4. The system shall automatically update the patient record with reserving a turn with the desired doctor.

Non-Functional Requirements (NFRs):

1- The system shall update and display the patient's turn within **5 seconds** of the booking confirmation.

2- All payment methods shall be saved even if the network connection is briefly lost.

3- All payment transactions and sensitive financial data shall be encrypted to prevent unauthorized access.

User Story #25
As a doctor, I want to set my weekly working hours and availability so patients can book appointments during those times.

Functional Requirements (FRs)
1- The system shall allow access to the schedule management page only for users with the "Doctor" role.
2-The system shall allow the doctor to create, edit, or delete their availability entries (days and time slots).
3- The system shall allow patients to book appointments only within the doctor's defined available slots.

Non-Functional Requirements (NFRs):

1- The system shall provide an intuitive and user-friendly interface for doctors to manage their schedules easily.
2- The system shall load or save a doctor's schedule within two seconds under normal conditions.
3- The system shall be compatible with major mobile devices.

WBS diagram

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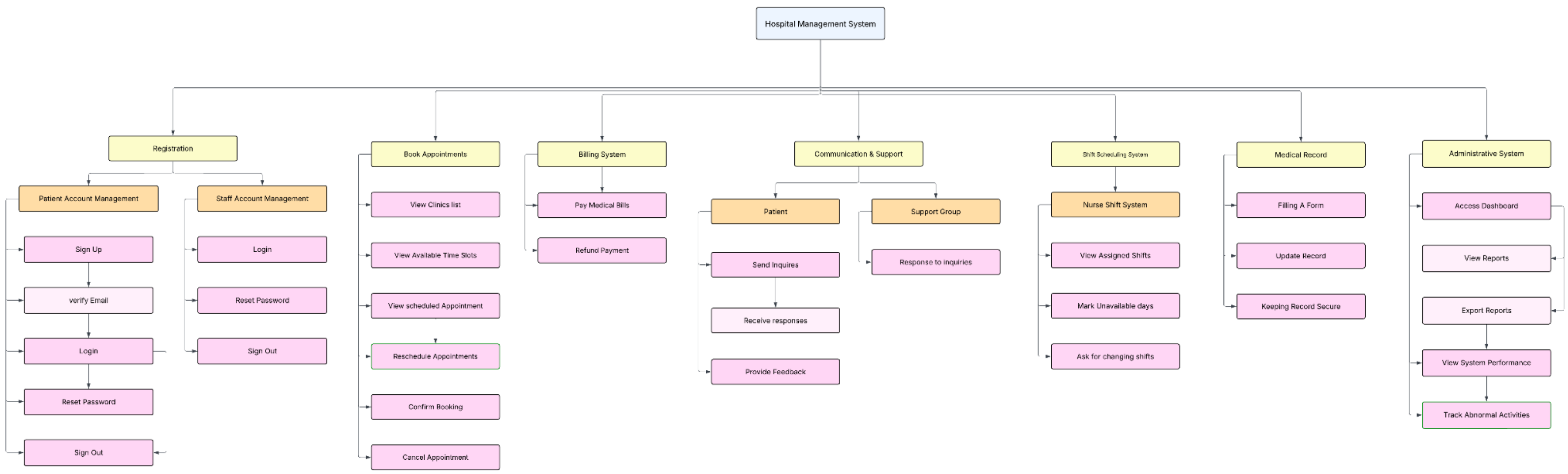

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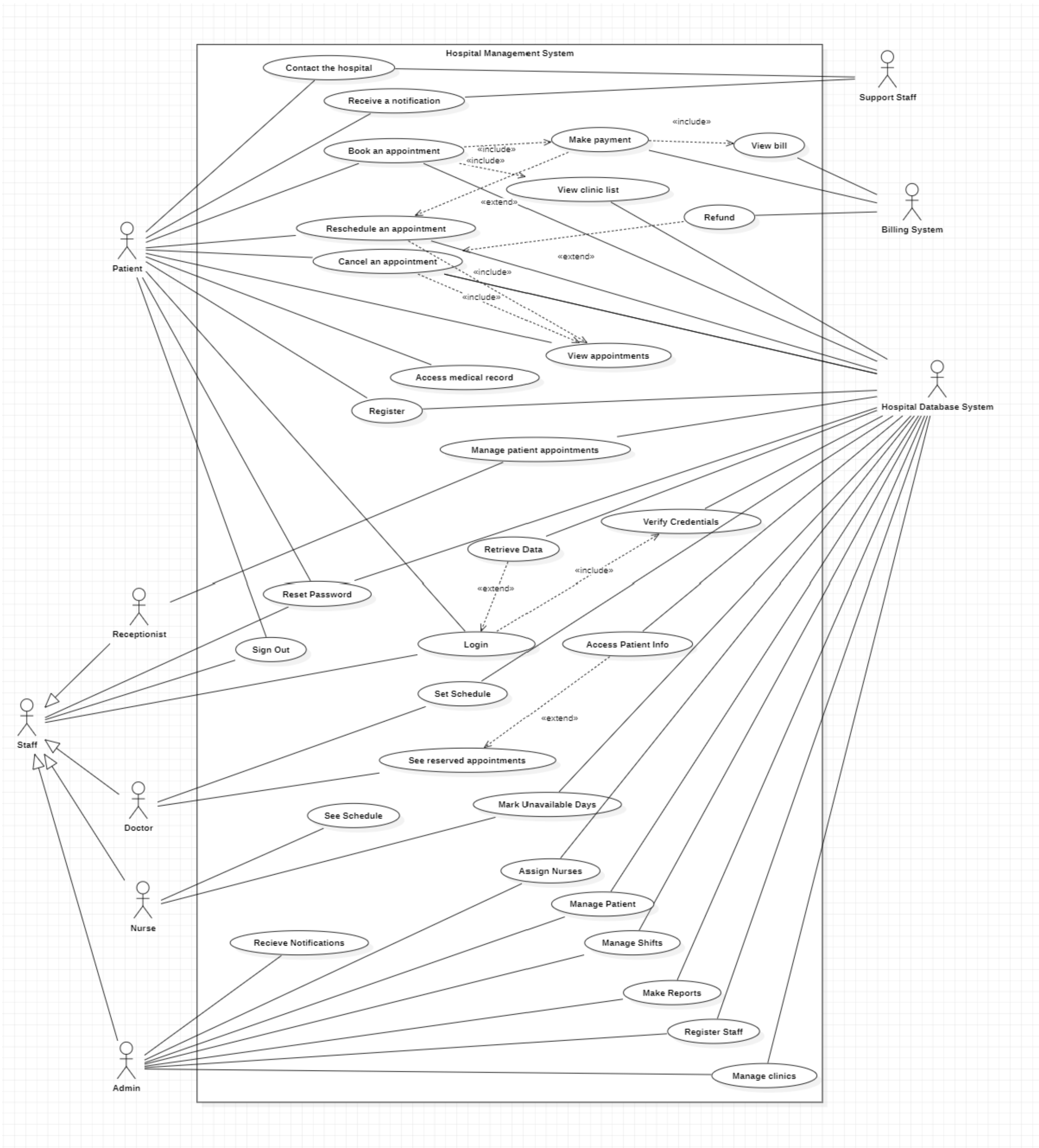
Use cases and Diagrams



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Risks Identified

1- Security and Data Protection Risks
- Unauthorized access to sensitive patient medical histories, diagnoses, and treatment records
- Authentication system vulnerabilities
- Inadequate multi-factor authentication for doctors, nurses, and administrative staff
- Insider Threats: Hospital staff misusing their legitimate access to patient records for unauthorized purposes

2- Technical Risks
- Poor information flow between different hospital departments using the system
- Notification System Failures: Missed appointment confirmations, schedule changes, or urgent medical alerts
- Slow Response Time if the user interface is complex loading and retrieving many analytics and data.
- Doctor schedules not properly updating across different system modules, causing appointment mismatches
- System allowing impossible scheduling scenarios (overlapping appointments, past-date bookings)
- Patient search functionality returning incorrect or incomplete results due to database problems

3- Clinical Safety Risks
- Inaccurate or incomplete medical records leading to wrong treatment decisions by doctors
- Incorrect prescription information display causing nurses to administer wrong medications or dosages
- System unavailability during critical patient care situations preventing access to vital medical information
- Poorly optimized search algorithms causing exponential performance degradation as data volume increases
- Application consuming increasing amounts of server memory over time, eventually causing system crashes

4- Staff Resistance and Training Issues
- Healthcare staff reluctance to transition from paper-based systems to digital HMS

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Core Features

User Story	Story Points	Priority
As a patient, I want to create a new account by providing my personal details and contact information so that I can access the hospital’s online services securely.	3	High
As a system user, I want to access a login page where I can enter my email and password so that I can securely sign in to my account.	2	High
As an admin, I want to create accounts for staff (doctors, nurses, receptionists) so that they can access the system.	3	High
As a patient, I want to book appointments so that I can visit my doctor at a convenient time.	3	High
As a patient, I want to view my upcoming and past appointments so that I can track my visit history.	2	Medium
As a patient, I want to reschedule my appointment so that I can change the time without losing my booking.	3	Medium
As a patient, I want to cancel my booked appointments so that I can free up the slot for others.	2	Medium
As a doctor, I want to view all appointments scheduled with me so that I can prepare accordingly.	2	High
As a receptionist, I want to view all appointments across doctors so that I can manage the clinic schedule.	2	High

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Document Links

- Requirements specification document : https://docs.google.com/document/d/1f1hmFtX_-Dvp_DHqrPH0mdQ6GPfLzlUduLcgVieGmNs/edit?tab=t.0#heading=h.dgz07mkfaeu2
- Design specification document : https://docs.google.com/document/d/1_nmkvvqNb9T8_X2EBdULvE-0o9T5U5Pq0KdWdFFvKXI/edit?tab=t.0
- Task Board Link : https://aucegypt-team-tk4cf28k.atlassian.net/jira/software/projects/HMS/boards/1/backlog